# PortFlow Container Terminal: Yard and Gate Flow

PORTFLOW · Supply Chain and Logistics Engineering · Strategy Consulting Hub

## Decision

Option B — Integrated flow package: shared visibility, appointments, AI-assisted yard planning — delivers 21.7% lower year-5 yard utilization (target 19.5%) for CAD $9,000,000 of the CAD $10,000,000 budget: NPV CAD $6,491,191 at 9.5%, payback 31.7 months. Five-year net financial impact (case formula): -CAD $16,500,000 (net savings).

## Why this option

Which package, within the $10M upfront budget, attacks PortFlow's core issue so that at +20% volume the yard runs at or below 85% utilization and trucks turn in under 45 minutes by year five?

| Opt | Alternative | Capex | Gain | In budget | Meets target | NPV | Payback | Score /10 |
|---|---|---|---|---|---|---|---|---|
| A | Gate quick win: truck appointments and release portal | CAD $4,100,000 | 10% | yes | no | CAD $4,157,625 | 28.5 months | 5.63 |
| B ✔ | Integrated flow package: shared visibility, appointments, AI-assisted yard planning | CAD $9,000,000 | 21.7% | yes | yes | CAD $6,491,191 | 31.7 months | 7.0 |
| C | Yard expansion on the 15-acre parcel with two added RTGs | CAD $14,500,000 | 14.3% | NO | no | -CAD $14,627,035 | > 60 months | 2.9 |

## Economics

| Capex | NPV @ WACC | IRR | Payback | Benefit/ Cost | Run-rate benefit | Levelized cost |
|---|---|---|---|---|---|---|
| CAD $9,000,000 | CAD $6,491,191 | 29.9% | 31.7 months | 1.46 | CAD $6,400,000/ yr | 128752.43 CAD per year per % point of improvement |

Bear case (performance -15%, capex +20%): NPV CAD $620,445, payback 45.2 months.

## Plan

- **Phase 1 (Months 1-3)** — Co-design and pilot, CAD $2,000,000. KPI: Pilot lanes: queue time -25%; release status shared within 1 hour; baseline rehandles measured
- **Phase 2 (Months 4-9)** — Scale appointments, positioning and AI advice, CAD $5,200,000. KPI: Dwell <= 5.3 days; truck turn <= 55 min; rehandles per box -20%
- **Phase 3 (Months 10-12)** — Stabilise, connect rail and customers, hand over, CAD $1,800,000. KPI: Dwell <= 4.8 days; truck turn < 45 min; utilization on track for <= 85% at year 5

## Top judge questions

**Q: Why not just build on the 15-acre parcel?**
Space alone fails the objective: 25,344 TEU / 28,000 TEU = 90.5% at year 5. It costs about $14.5M, over the $10M budget, and must be phased around live operations. Cutting dwell to 4.7 days reaches about 83% on today's footprint. We keep the parcel as a phase-2 option if dwell stays above 5.6 days.

**Q: What if dwell does not fall as much as you assume?**
The threshold is explicit: dwell at or below 4.83 days meets 85% without land. If dwell is above 5.3 days at month 9 (our gate trigger), we tighten the free-time tariff and start the parcel design. With dwell at 5.6 days or less, the parcel would then meet the objective.

**Q: How confident are you in the $6.4M of annual savings?**
Each saving has an owner and a mechanism. Break-even needs only $3.1M a year. If penalties fall 25% instead of 50%, the five-year impact is still -$6.5M.